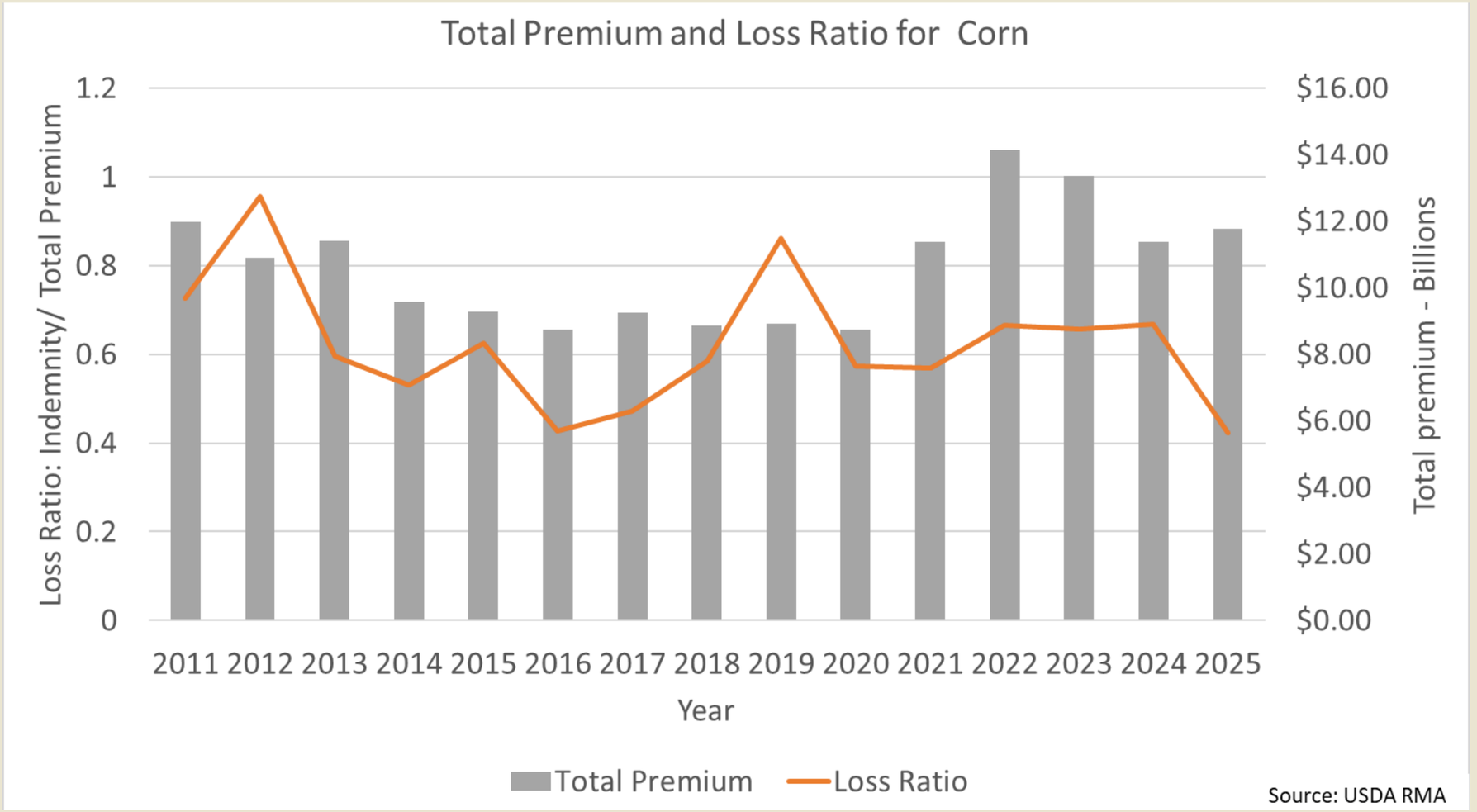

CropMIND
Multi-Agent
Automated Corn Disaster Decision-Support and Claims
Vision
Agronomy
Legal

Current System is Failing Farmers



Loss Ratio: Indemnity/ Total Premium

CropMIND

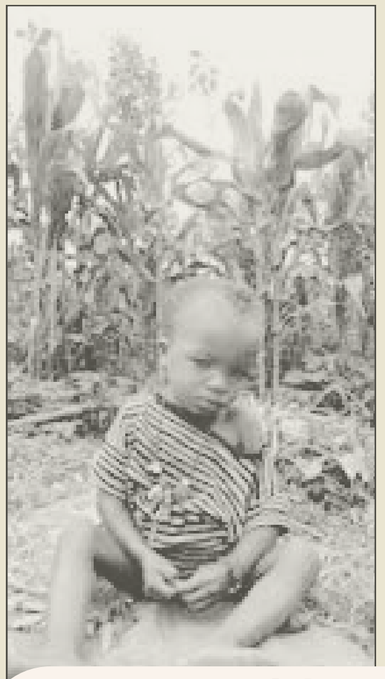
The Daily Cost of Delay

In Illinois, every 24 hours of waiting in June costs a 500-acre farm $7,000











Seasonal Crisis

April through June is the peak season for severe convective weather and flooding disasters in IL

Claims Latency:

Farmers typically have to wait 7 to 14 days to meet with an insurance adjuster.

CropMIND

End to End Pipeline

input

- Farmer-Generated Evidence
- Geographical & Topographic Context
- Soil & Environmental Parameters

processing

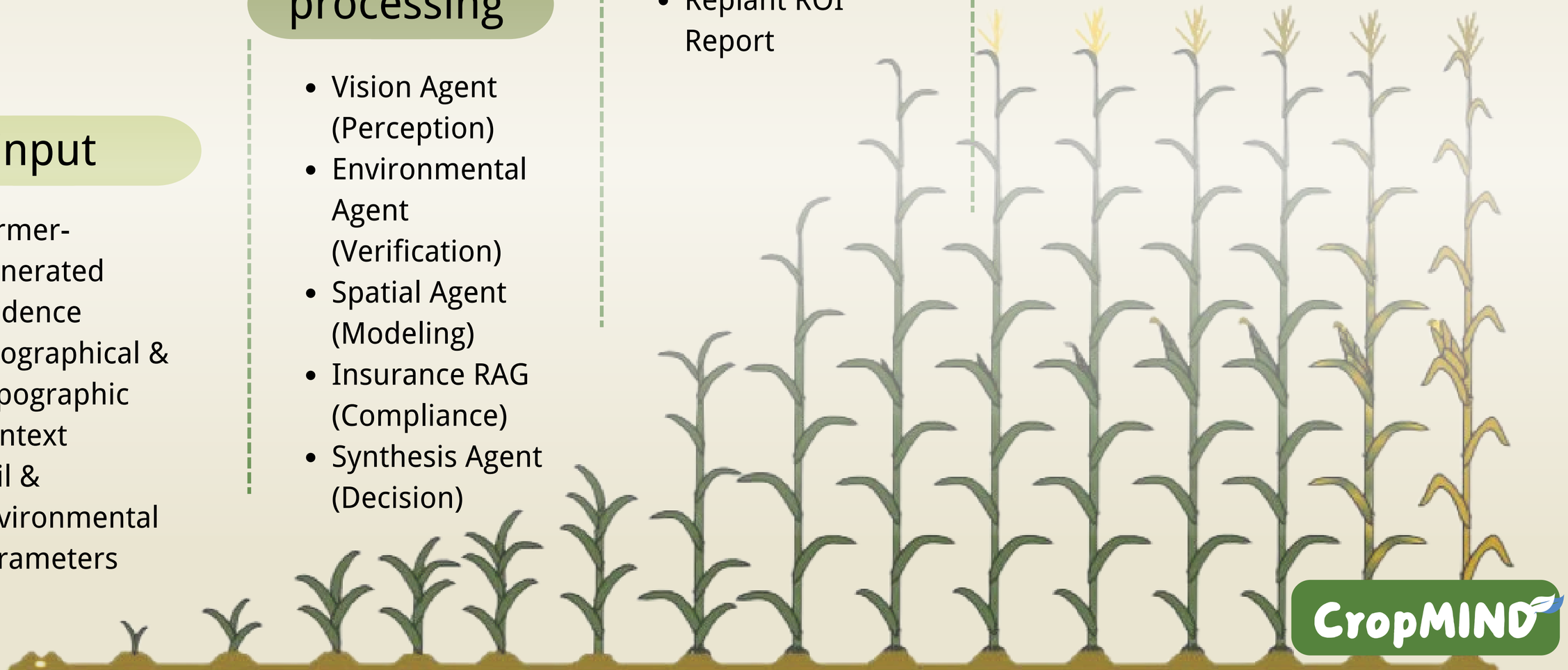
- Vision Agent (Perception)
- Environmental Agent (Verification)
- Spatial Agent (Modeling)
- Insurance RAG (Compliance)
- Synthesis Agent (Decision)

output

- 3D Field Risk Map
- Legal to-do List
- Replant ROI Report

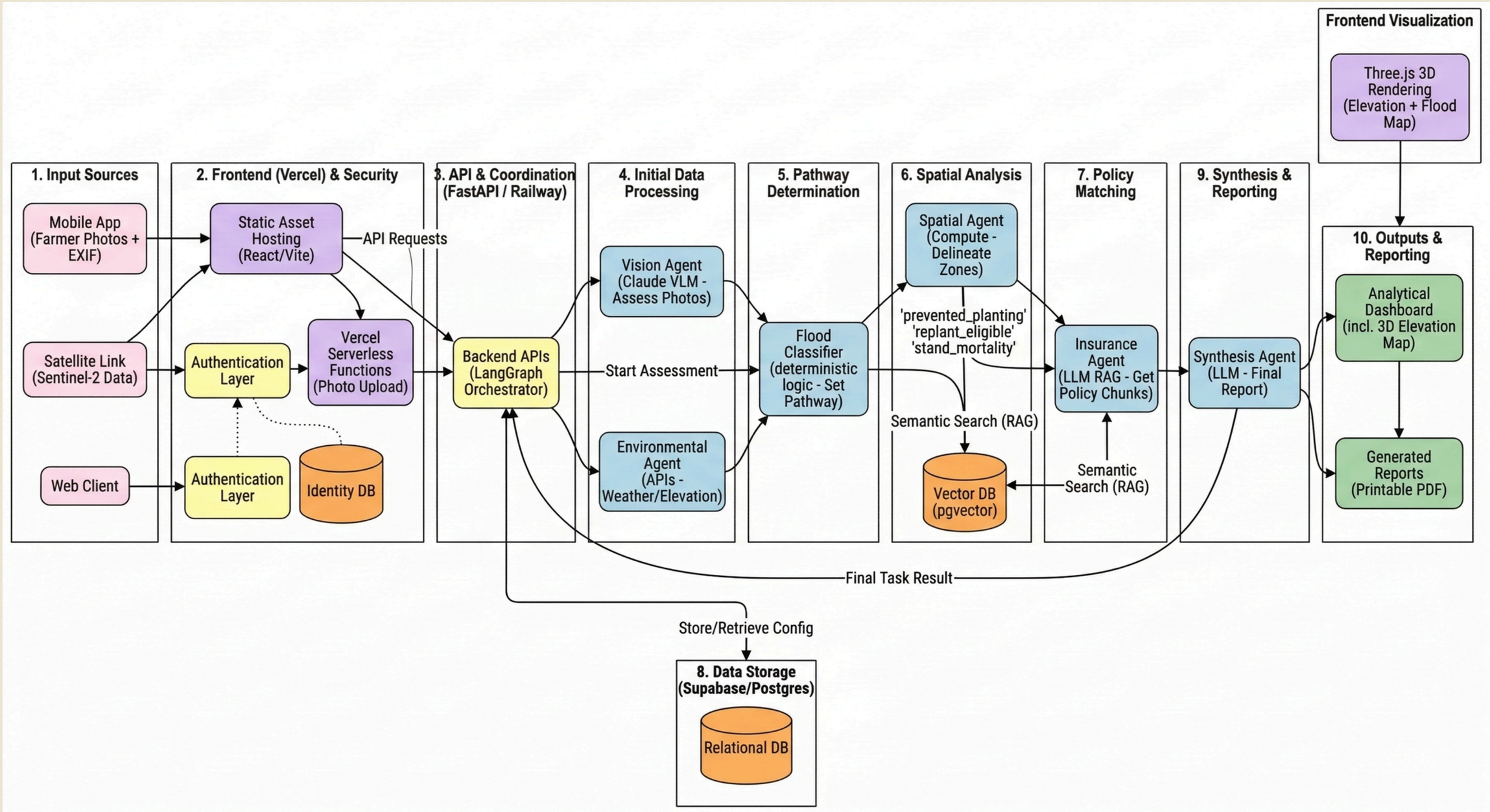
feedback

- User feedback to optimize future analyzation

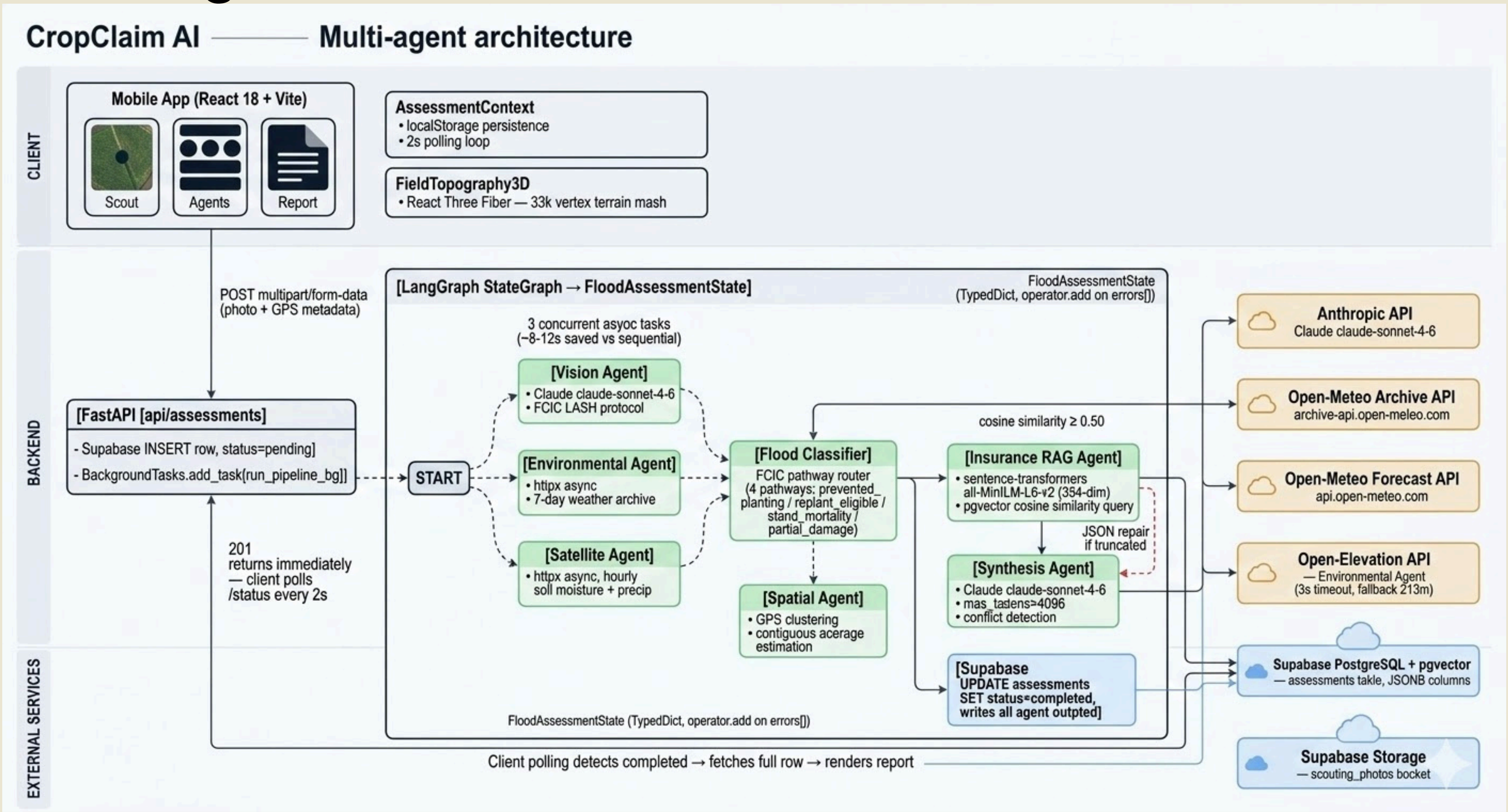


CropMIND

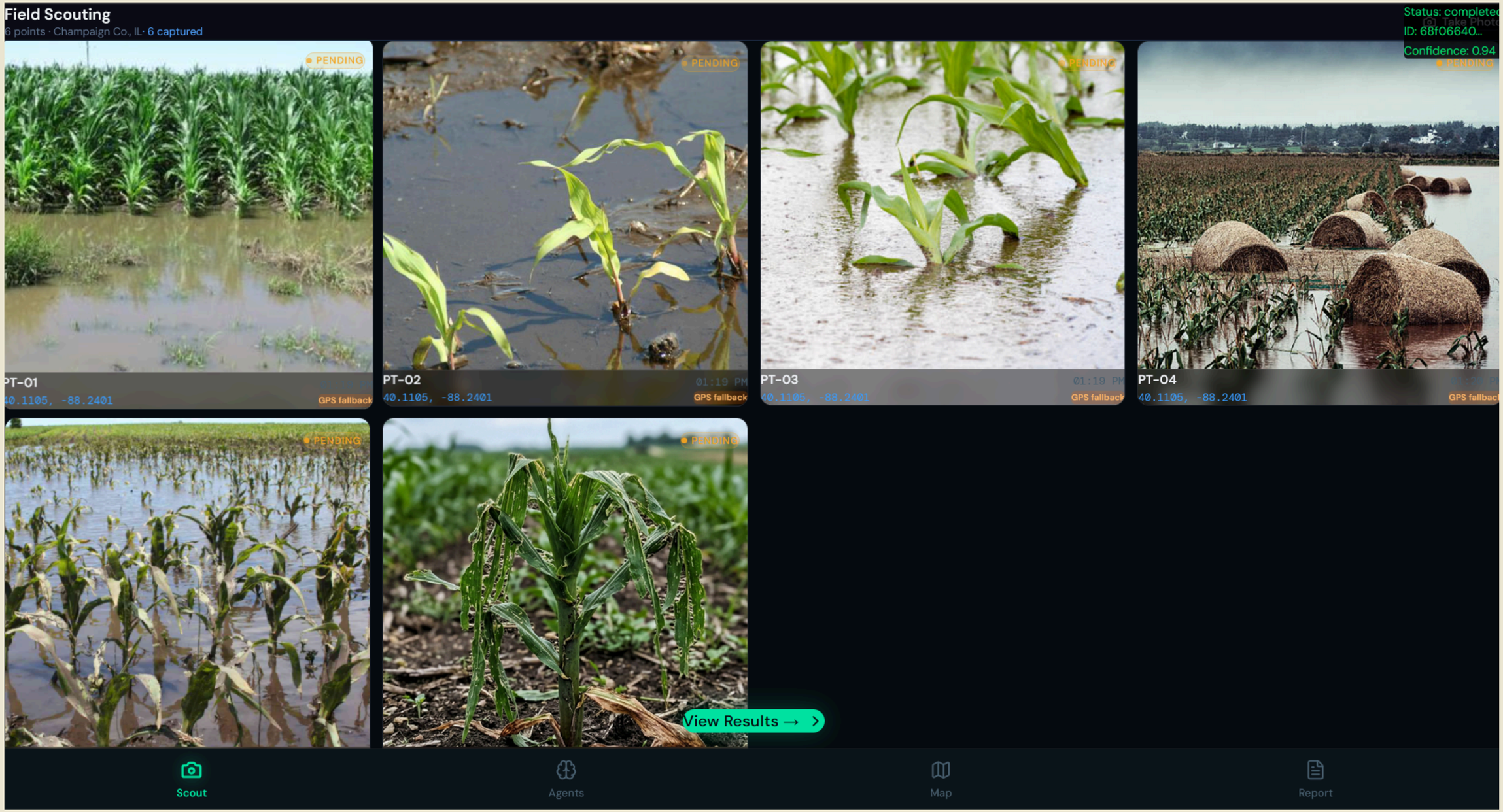
Techical Architecture



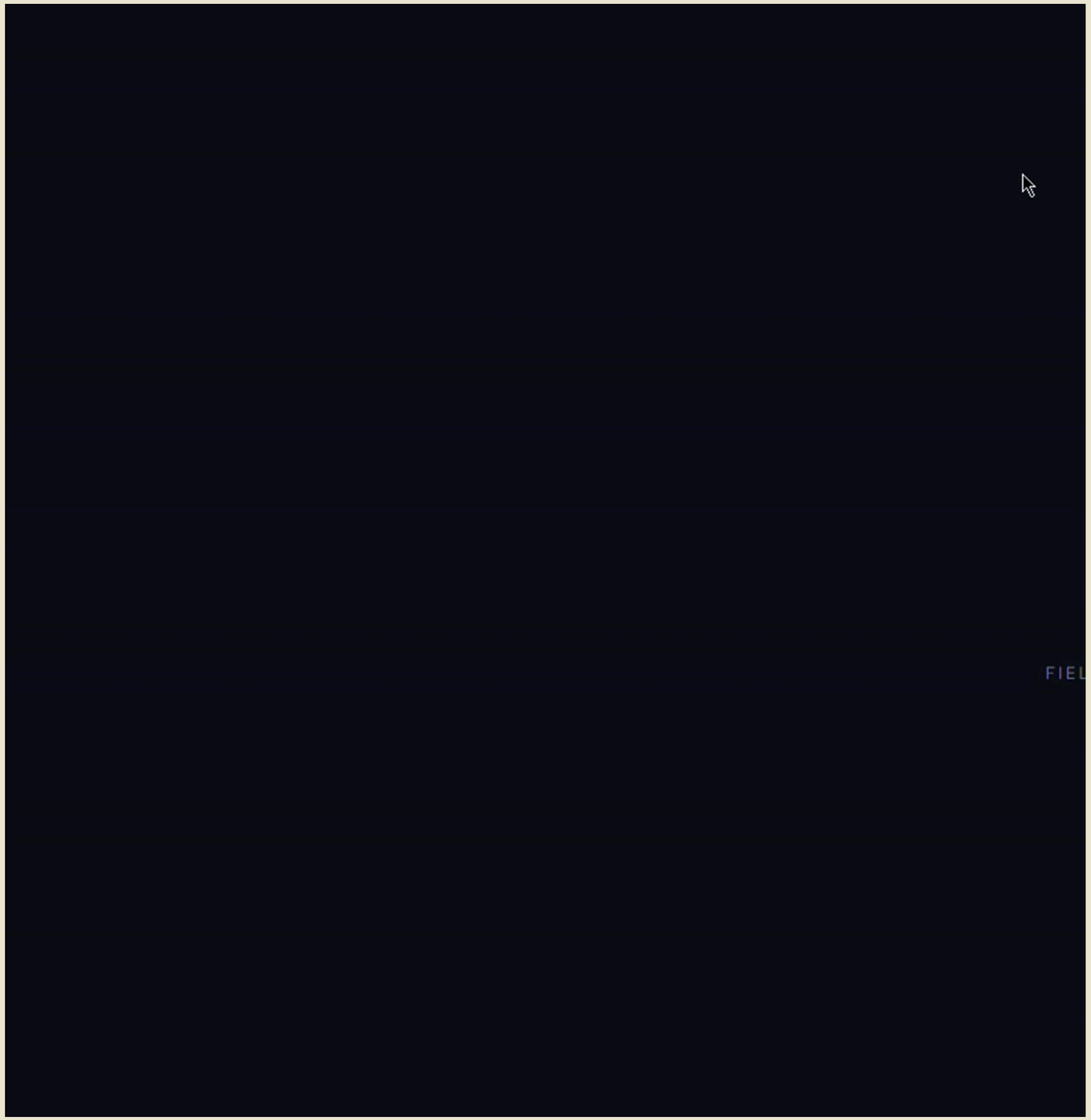
Multi agent architecture



Field Scout


Field Scouting
6 points · Champaign Co., IL · 6 captured
Status: completed
ID: 68f06640...
Confidence: 0.94
PENDING
PT-01
40.1105, -88.2401
GPS fallback
PT-02
01:19 PM
40.1105, -88.2401
GPS fallback
PT-03
01:19 PM
40.1105, -88.2401
GPS fallback
PT-04
40.1105, -88.2401
GPS fallback
View Results →
Scout
Agents
Map
Report

Agent Council



VISION AND ENVIRONMENTAL INTELLIGENCE

AGENT 1: VISION

METHODOLOGY

- ✓ FCIC loss adjuster persona (LASH protocol)
- ✓ Horizontal leaf method for damage assessment
- ✓ Claude 3.5 Sonnet visual reasoning

OUTPUT SCHEMA

```
VisionResult {
  crop_type: str,
  growth_stage: str,
  damage_types: list[str],
  defoliation_percent: float,
  stand_loss_percent: float,
  confidence: float
}
```

AGENT 2 & 3: ENVIRONMENTAL

WEATHER INTELLIGENCE

- Open-Meteo Archive: 7-day precip sum
- Open-Elevation: Terrain-aware risk scoring
- Soil moisture monitoring (0-7cm depth)

RISK SCORING HEURISTIC

- Elevation < 220m AND Precip > 50.8mm
- → Classified as "High Flood Risk"
- Satellite Agent: Hourly soil moisture forecast

FLOOD CLASSIFICATION & SPATIAL ANALYSIS

4 FCIC PATHWAYS

PREVENTED PLANTING

Event occurs before final planting date; requires 20-acre threshold check.

REPLANT ELIGIBLE

Event within 25 days of planting; survival rates dictate replant necessity.

STAND MORTALITY

Survival < 50% at V3-V5 stage; high-confidence loss identification.

PARTIAL DAMAGE

Survival ≥ 50% in late-season; focus on yield impact estimation.

SPATIAL AGENT LOGIC

GPS Clustering Algorithm

- Elevation-based zone assignment for contiguous flood checks.
- Submerged points (survival <20%) clustered by proximity.
- Acreage estimation: cluster_count × spacing² × 0.000247.

3D Digital Twin (React Three Fiber)

- 180×180 mesh grid with 32,400 quads.
- Vertex color updates in ~0.8ms per threshold change.
- World scale: 1WU ≈ 44.4m lat, 34m lng (~149 acres total).

INSURANCE RAG: VECTOR RETRIEVAL PIPELINE

RETRIEVAL PIPELINE

EMBEDDING MODEL

all-MiniLM-L6-v2 (384-dim, ~80MB)
Loaded once at import, cached in memory for sub-ms inference.

QUERY CONSTRUCTION

Dynamic prompt engineering based on **Flood Pathway** and **Crop Type** to ensure high-relevance matching.

```
# Supabase pgvector RPC
match_policies(
  query_embedding,
  match_threshold: 0.5,
  match_count: 2
)
```

TECHNICAL SPECS

VECTOR DATABASE

Supabase pgvector 0.7
IVFFlat indexing (lists=100) for efficient Approximate Nearest Neighbor (ANN) search.

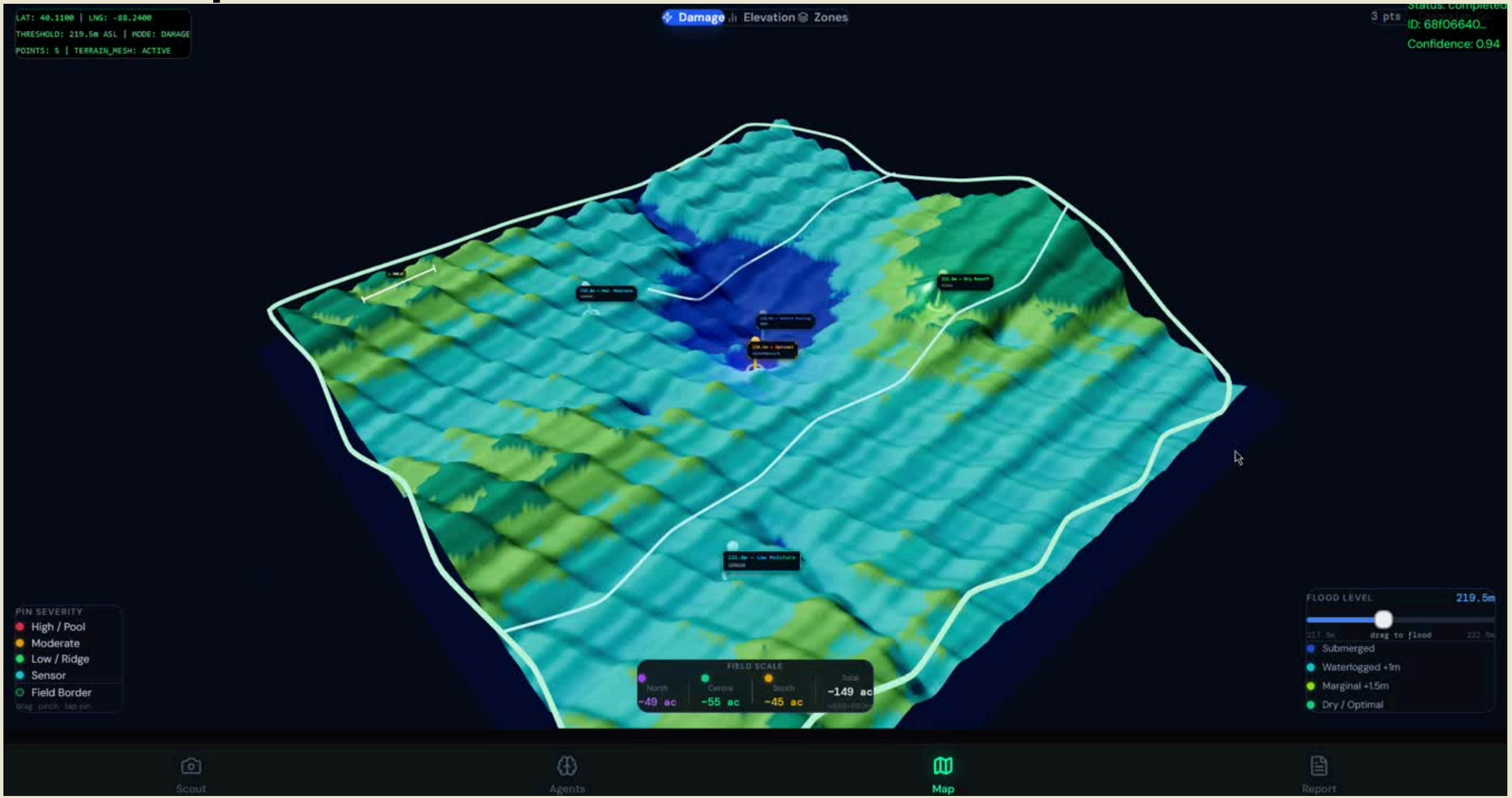
ACTION ITEM GENERATION

- ✓ **Universal:** File Notice of Loss within 72h.
- ✓ **Grounded:** Extracted from policy chunks.
- ✓ **Pathway-Specific:** e.g., "Do not replant".

COMPLIANCE

Every recommendation is grounded in official **FCIC Policy Documentation** via retrieved text snippets.

3D Map


LAT: 40.1100 | LNG: -88.2400
THRESHOLD: 219.5m ASL | MODE: DAMAGE
POINTS: 5 | TERRAIN_MESH: ACTIVE
Damage
Elevation
Zones
3 pts
ID: 68f06640...
Confidence: 0.94
PIN SEVERITY
High / Pool
Moderate
Low / Ridge
Sensor
Field Border
FIELD SCALE
North
~49 ac
Centre
~55 ac
South
~45 ac
Total
~149 ac
FLOOD LEVEL
219.5m
drag to flood
Submerged
Waterlogged +1m
Marginal +1.5m
Dry / Optimal
Scout
Agents
Map
Report

3D DIGITAL TWIN: FIELD-SCALE VISUALIZATION

TERRAIN GENERATION

Procedural Heightmap

Deterministic **computeHeight(gx, gy)** function using 3 octaves of sine/cosine noise.

Geometry Specs

180×180 segments yielding **32,400 quads** and 33,124 vertices for high-fidelity terrain.

Features

Encoded central basins, drainage gullies, and secondary pools for realistic flood simulation.

COORDINATE SYSTEM

GPS to World Mapping

Precise conversion: **1WU ≈ 44.4m lat, 34m lng**. Sign flip on Z-axis is critical for orientation.

```
wx = (lng - LNG_CTR) * 2500
wz = -(lat - LAT_CTR) * 2500
```

Field Borders

Dynamic tubes that **hug the terrain** by sampling heightAt() at each control point.

PERFORMANCE & MODES

Real-Time Interaction

Vertex color updates in **<1ms on M1**, ensuring smooth flood threshold slider response.

View Modes

- **Damage:** 4 flood zones.
- **Elevation:** 5 hypsometric bands.
- **Zones:** Geographical sectors.

Damage Dashboard

CLM-2026-0314 · 2026-03-08T15:55:50.095+00:00 · Field Unit 004

Status: completed
ID: 68f06640...
Confidence: 0.94

DAMAGE BY ZONE

2 zones

Zone A — Ridge

SEVERE

Flood Inundation 2 scout pts

EST. YIELD LOSS ~45%

Field notes

Zone B — Depression

LOW

Wind Lodging 1 scout pts

EST. YIELD LOSS ~3%

Field notes

FCIC POLICY MATCHES

2 matched

Coarse Grains Crop Provisions Section 8 (b)(1) — Triggered

Replant payment is authorized if the acreage replanted is at least 20 acres or 20 percent of the insured planted acreage for the unit.

Basic Provisions Section 14 (a)(2) — Triggered

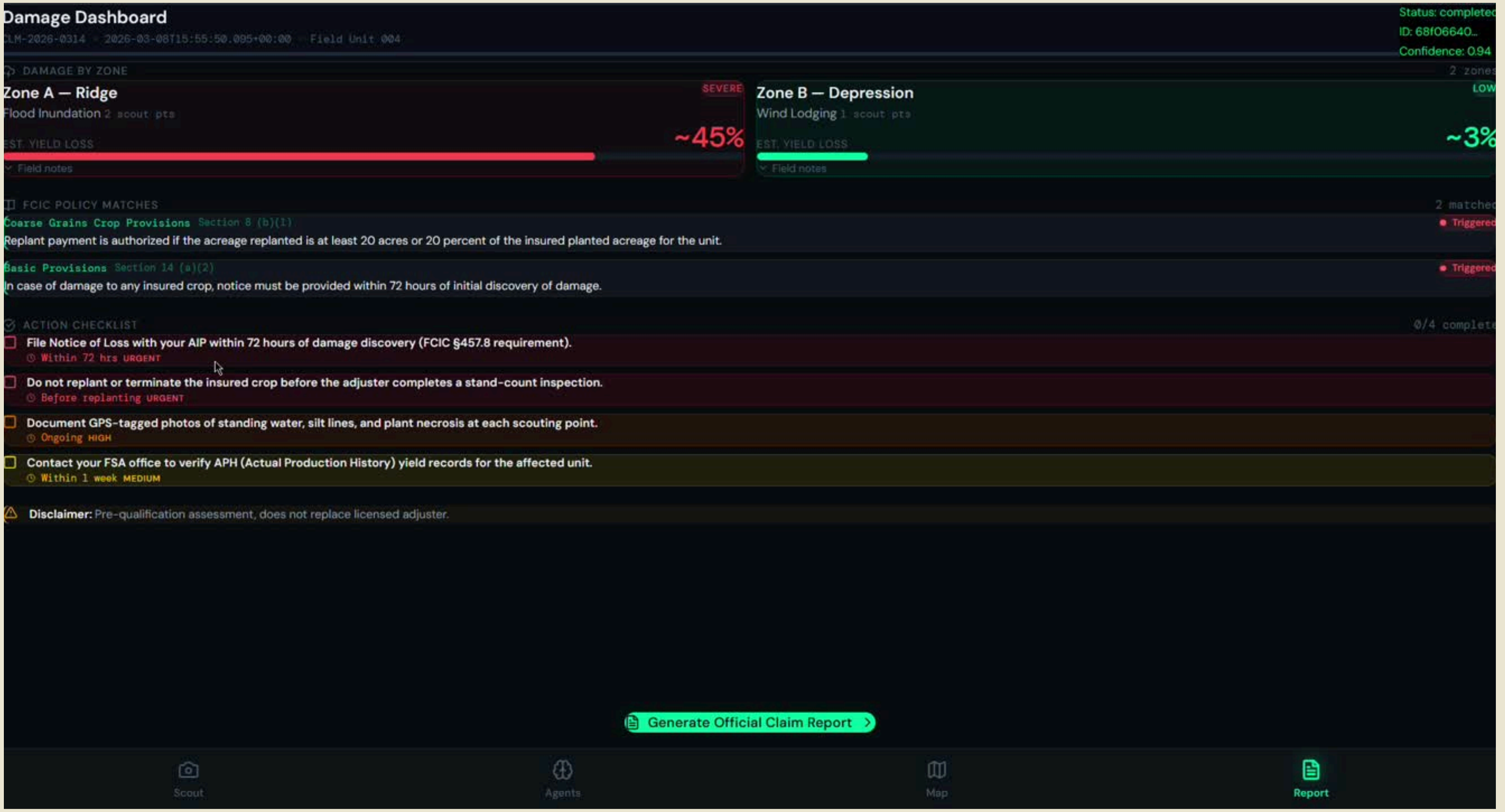
In case of damage to any insured crop, notice must be provided within 72 hours of initial discovery of damage.

ACTION CHECKLIST

0/3 complete

- [ ] **Review Replant Payment calculation per Section 8.**
  See report HIGH
- [ ] **Dispatch adjuster to verify exact acreage of the 22-acre contiguous cluster.**
  See report HIGH
- [ ] **Verify seed receipts for original planting date.**
  See report HIGH

Disclaimer: Pre-qualification assessment, does not replace licensed adjuster.


Damage Dashboard
CLM-2026-0314 · 2026-03-08T15:55:50.095+00:00 · Field Unit 004
Status: completed
ID: 68f06640...
Confidence: 0.94
DAMAGE BY ZONE
2 zones
Zone A — Ridge
SEVERE
Flood Inundation 2 scout pts
EST. YIELD LOSS
~45%
Field notes
Zone B — Depression
LOW
Wind Lodging 1 scout pts
EST. YIELD LOSS
~3%
Field notes
FCIC POLICY MATCHES
2 matched
Coarse Grains Crop Provisions Section 8 (b)(1)
Triggered
Replant payment is authorized if the acreage replanted is at least 20 acres or 20 percent of the insured planted acreage for the unit.
Basic Provisions Section 14 (a)(2)
Triggered
In case of damage to any insured crop, notice must be provided within 72 hours of initial discovery of damage.
ACTION CHECKLIST
0/4 complete
File Notice of Loss with your AIP within 72 hours of damage discovery (FCIC §457.8 requirement).
Within 72 hrs URGENT
Do not replant or terminate the insured crop before the adjuster completes a stand-count inspection.
Before replanting URGENT
Document GPS-tagged photos of standing water, silt lines, and plant necrosis at each scouting point.
Ongoing HIGH
Contact your FSA office to verify APH (Actual Production History) yield records for the affected unit.
Within 1 week MEDIUM
Disclaimer: Pre-qualification assessment, does not replace licensed adjuster.
Generate Official Claim Report
Scout
Agents
Map
Report

Challenges

EDGE CONNECTIVITY & PAYLOAD COMPRESSION

THE RURAL CHALLENGE

BANDWIDTH CONSTRAINTS

Farmers upload **12MP+ field evidence** from rural areas with unstable **3G/LTE connections**, causing frequent timeouts.

TOKEN OVERHEAD

Massive image payloads lead to excessive **VLM token costs** and increased processing latency.

THE MITIGATION

CLIENT-SIDE PROCESSING

Implement **WebP compression** and **adaptive downsampling** on the mobile client before the network request.

CRYPTOGRAPHIC SIGNING

We extract and **cryptographically sign** the EXIF/GPS metadata first to ensure auditability before compression.

10MB → 300KB 97% REDUCTION

**Maintains full agronomic fidelity required by the Vision Agent.*

Q&A









CropMIND

Team Members